Causality Assessment for Case ADR-2026-014

Suspected drug: Allopurinol 300 mg OD | Reaction: DRESS syndrome

1. WHO-UMC Causality Assessment

Criterion	Finding in this case	Met?
Plausible time relationship	Onset on Day 21 to 25 after starting the drug, which fits the 2 to 8 week latency of DRESS.	Yes
Unlikely to be explained by disease or other drugs	Blood cultures and viral hepatitis markers were negative. Long-term drugs were stable for years, colchicine was stopped 7 days before onset, and paracetamol started after the rash.	Yes
Response to withdrawal clinically reasonable	Improved over 10 days, and LFTs and creatinine normalized by week 6. Steroids were given at the same time, so this is a mild confounder.	Yes
Event definitive pharmacologically / phenomenologically	DRESS with allopurinol is a well-recognized, characteristic reaction.	Yes
Rechallenge positive	Not done, and contraindicated.	No

Why not Certain?

Certain requires a positive rechallenge, or an event that is definitive pharmacologically, with no other explanation. No rechallenge was done, and it would be unsafe.

Why not Possible?

Possible applies when the reaction could also be explained by disease or other drugs, or when information on withdrawal is missing. Here, alternatives were reasonably excluded and dechallenge is documented.

WHO-UMC result: Probable / Likely

2. Naranjo Algorithm

Sr.no.	Question	Answer	Score
1	Previous conclusive reports of this reaction?	Yes (allopurinol-DRESS is well documented)	+1
2	Did the event appear after the drug was given?	Yes	+2
3	Did it improve on discontinuation?	Yes	+1
4	Did it reappear on readministration?	Not done	0
5	Alternative causes that could on their own explain it?	No	+2
6	Reaction with placebo?	Not done	0
7	Drug detected at toxic concentration?	Not measured	0
8	Dose-response relationship?	Not known	0
9	Similar reaction to the same or similar drug before?	No	0
10	Confirmed by objective evidence?	Yes (eosinophilia 1,400/uL, ALT 210, creatinine 1.9)	+1
	TOTAL		7

Naranjo interpretation: >=9 Definite, 5 to 8 Probable, 1 to 4 Possible, <=0 Doubtful. A score of **7** means **Probable.**

3. Supporting Tool: RegiSCAR DRESS Validation Score

This tool confirms the diagnosis rather than causality, but it strengthens the case that the reaction is truly DRESS.

Item	Finding	Score
Fever >=38.5 C	38.6 C (101.4 F)	0
Enlarged lymph nodes	Cervical and axillary	+1
Eosinophilia	1.4 x 10^9/L (0.7 to 1.499 range)	+1
Atypical lymphocytes	Not reported	0
Rash >50% of body surface	Diffuse, trunk and limbs (assumed)	+1
Rash features suggestive of DRESS	Facial edema, maculopapular	+1
Skin biopsy suggesting DRESS	Not done	0
Organ involvement	Liver and kidney (two organs)	+2
Resolution >=15 days	Yes	0
Other causes excluded (>=3 tests)	Only partially tested	0
TOTAL		6

A score above 5 is a definite DRESS case. If the rash extent was under 50% of body surface, the total would be **5**, which is still **Probable.**

4. Drug-by-Drug Causality Summary

Drug	WHO-UMC category	Reasoning
Allopurinol	Probable	Fits timing, characteristic reaction, positive dechallenge
Colchicine	Unlikely	Stopped Day 14, 7 days before onset, and not a typical DRESS trigger
Atorvastatin, amlodipine, metformin	Unlikely	Years of uneventful use
Paracetamol	Unlikely	Started after the rash began (it may have been taken for the fever)

5. Final Conclusion

Allopurinol is the probable cause of DRESS in this patient (WHO-UMC: Probable; Naranjo: 7).

What would change the category

A positive rechallenge would make it Certain, but this is never done for DRESS because it is life-threatening.
A competing explanation, such as a viral reactivation not yet tested for (for example HHV-6 or EBV), or a second new drug started close to onset, would lower it toward Possible.